

DANVILLE, VA, USA

WMO: 724106

Lat: 36.573N

Lon: 79.335W

Elev: 174

StdP: 99.25

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13728

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-5.8	-17.6	0.8	-3.5	-15.1	1.0	-2.2	9.9	7.6	8.7	7.7	2.1	260	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.1	23.6	32.9	23.5	31.9	23.2	25.5	31.4	24.9	30.5	24.3	29.5	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.1	27.9	23.2	18.3	27.1	22.7	17.8	26.6	78.9	31.3	76.3	30.8	73.9	29.6	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 7.3	(c) 6.2	(e) -11.9	(f) 36.1	(g) 2.9	(h) 1.8	(i) -14.0	(j) 37.4	(k) -15.7	(l) 38.4	(m) -17.4	(n) 39.4	(o) -19.5	(p) 40.7		
(a) 8.3	(b) 7.3	(c) 6.2	(e) -12.7	(f) 27.0	(g) 2.8	(h) 1.0	(i) -14.7	(j) 27.7	(k) -16.3	(l) 28.3	(m) -17.8	(n) 28.8	(o) -19.8	(p) 29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	4.2	5.7	9.7	14.9	19.5	23.7	25.6	25.0	21.4	15.4	9.4	5.6		
	DBStd	8.75	5.50	5.17	5.30	4.53	3.86	2.89	2.29	2.45	3.34	4.41	4.70	4.93		
	HDD10.0	647	196	139	73	10	0	0	0	0	0	7	67	154		
	HDD18.3	2010	439	354	272	121	35	2	0	0	11	112	269	395		
	CDD10.0	2492	15	18	62	157	294	409	483	467	343	174	49	18		
	CDD18.3	814	0	0	3	18	71	162	225	208	104	21	1	1		
	CDH23.3	7386	1	3	39	224	650	1488	2083	1869	821	191	13	3		
	CDH26.7	2903	0	0	4	50	198	597	918	802	291	44	1	0		
(14) Wind	WSAvg	2.6	2.9	3.0	3.2	3.2	2.6	2.3	2.1	2.0	2.2	2.3	2.5	2.5		
(15) Precipitation	PrecAvg	1095	91	84	101	87	100	94	110	96	97	85	76	80		
	PrecMax	1607	222	168	265	210	196	371	224	245	372	276	194	163		
	PrecMin	728	18	10	17	8	29	29	41	22	5	3	7	8		
	PrecStd	191	46	38	54	51	46	69	41	50	79	60	42	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	23.0	26.9	30.9	32.5	35.0	36.3	36.0	33.9	30.5	25.2	22.6		
		MCWB	15.8	15.5	16.3	18.8	22.3	23.9	24.3	23.7	21.9	21.7	16.8	15.8		
	2%	DB	18.4	20.1	24.1	27.9	30.7	33.0	34.1	34.0	31.9	27.5	22.4	19.0		
		MCWB	14.4	13.8	15.0	17.7	21.5	23.5	24.1	23.8	22.2	19.4	15.7	15.2		
	5%	DB	16.2	17.6	21.8	26.2	29.0	31.8	32.8	32.5	30.1	25.4	20.2	16.7		
		MCWB	12.2	11.9	13.6	16.8	21.0	23.1	24.0	23.5	21.5	18.6	14.9	13.2		
	10%	DB	13.2	14.8	19.0	23.7	27.3	30.2	31.4	31.1	27.9	23.0	17.9	13.9		
		MCWB	9.2	9.5	12.5	15.8	19.9	22.5	23.6	23.2	20.8	17.8	13.6	10.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	17.8	18.2	20.8	24.0	25.5	26.3	26.2	24.8	23.5	19.6	18.5		
		MCDB	19.4	21.0	23.6	25.3	29.3	31.7	32.6	32.2	29.5	26.4	22.1	21.4		
	2%	WB	15.3	15.6	16.8	19.4	22.8	24.7	25.6	25.3	23.8	21.7	17.9	16.2		
		MCDB	18.1	19.2	21.6	25.2	28.3	30.7	31.7	31.1	28.5	25.4	20.2	17.9		
	5%	WB	12.9	13.0	15.4	18.3	21.9	24.0	25.1	24.7	22.9	20.3	16.3	13.7		
		MCDB	15.2	16.0	19.9	23.8	27.3	29.7	30.9	30.0	27.3	23.7	19.5	16.0		
	10%	WB	9.7	10.2	13.8	17.1	20.9	23.3	24.4	24.0	22.1	18.8	14.3	10.8		
		MCDB	12.6	13.9	17.9	22.0	25.3	28.6	29.7	28.8	26.0	22.0	17.0	13.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.5	12.4	13.1	12.0	11.4	10.8	10.9	11.1	12.5	12.5	10.6		
		MCDBR	14.1	15.2	15.6	15.8	14.0	13.1	12.7	13.0	13.9	14.7	15.1	13.5		
	5% WB	MCWBR	9.9	10.1	8.4	7.4	6.3	5.2	4.6	4.5	5.5	7.5	9.2	9.6		
		MCWBR	12.5	13.0	12.8	13.0	11.9	11.5	11.1	10.9	10.7	11.3	11.5	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.322	0.352	0.395	0.435	0.481	0.518	0.517	0.408	0.367	0.322	0.307		
	taud		2.511	2.456	2.414	2.304	2.214	2.128	2.047	2.027	2.330	2.428	2.514	2.535		
	Ebn at Noon		896	917	914	888	853	811	777	767	844	852	865	867		
	Edh at Noon		80	95	109	128	143	155	167	168	117	97	79	74		
(44) All-Sky Solar Radiation	RadAvg		2.48	3.17	4.20	5.26	5.83	6.32	5.90	5.37	4.43	3.60	2.73	2.12		
	RadStd		0.17	0.38	0.29	0.42	0.48	0.46	0.39	0.28	0.42	0.44	0.21	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	+85

Nomenclature: See separate page